EAGGL Methods (draft)

1 Methods

1.1 Overview: a mechanistic model for a single trait anchor

EAGGL (Enrichment Analysis Gene Grouping and LLMs) aims to infer latent molecular mechanisms for a single anchored trait, starting from: (i) a gene-by-annotation matrix (typically gene sets) describing functional context, and (ii) PIGEAN outputs that quantify genetic support for genes and annotation effects for the anchor trait.

Observed quantities. Let genes be indexed by $i \in \{1, \ldots, G\}$ and gene annotations (including gene sets) by $j \in \{1, \ldots, A\}$. Let $X_{ij} \geq 0$ denote the value of annotation j for gene i. In the most common case, $X_{ij} \in \{0, 1\}$ indicates membership of gene i in gene set j, but EAGGL supports non-binary weights (e.g. continuous annotations that are thresholded/capped upstream).

For the anchor trait, PIGEAN provides gene-level and annotation-level evidence summaries. EAGGL consumes gene-level evidence summaries (e.g. combined support and related quantities from PIGEAN) and annotation/gene-set summaries (e.g. gene-set effects such as `betas` or `betas_uncorrected` in the EAGGL runtime state). To obtain non-negative weights for factorization, the current code maps these real-valued summaries to $[0, 1]$ via a logistic transform with a background prior $\pi_0 \in (0, 1)$:

$$y_i := \text{gene evidence summary for gene } i \tag{1}$$

$$b_j := \text{gene-set/annotation summary for annotation } j \tag{2}$$

$$p_i := \text{sigmoid}(y_i + \text{logit}(\pi_0)) \tag{3}$$

$$q_j := \text{sigmoid}(b_j + \text{logit}(\pi_0)) \,, \tag{4}$$

where p_i and q_j are used as *anchor weights* throughout EAGGL (including matrix weighting, filtering, and post-processing). where d_i is a binary indicator that gene i is trait-relevant, a_j is an analogous indicator that annotation j is trait-relevant, and D denotes the genetic evidence. In the current codebase, π_0 is set via `--background-prior`, and the logistic transformation is applied to whichever gene and gene-set evidence quantities are available at runtime.

Latent mechanisms. We assume K latent molecular mechanisms indexed by $k \in \{1, \ldots, K\}$. Each mechanism is represented by two non-negative loading matrices:

$$f^g_{ik} \in [0, 1] \quad (\text{gene } i \text{ participates in mechanism } k) \tag{5}$$

$$f^a_{jk} \in [0, 1] \quad (\text{annotation } j \text{ captures mechanism } k). \tag{6}$$

Under simplifying assumptions (including independence approximations and the interpretation of p_i and q_j as posterior probabilities), the model used to motivate EAGGL asserts that a weighted

gene-by-annotation matrix is approximately low-rank:

$$p_i\, q_j\, X_{ij} \;\approx\; \sum_{k=1}^{K} f^g_{ik} f^a_{jk} \;+\; \epsilon_{ij}, \tag{7}$$

where ϵ_{ij} is a residual term. Because all terms are non-negative, Eq. 7 suggests non-negative matrix factorization as a natural approximation to the mechanistic decomposition.

1.2 Bayesian NMF approximation to the single-anchor model

Matrix orientation and weights. In the EAGGL implementation, the factorization is performed on a gene-set-by-gene matrix

$$V \in \mathbb{R}_+^{N\times M}, \qquad V_{ji} := X_{ij}, \tag{8}$$

where $N = A$ gene sets/annotations and $M = G$ genes. The anchor trait induces non-negative weights on rows and columns:

$$\boldsymbol{q} \in [0,1]^N, \quad q_j \equiv q_j(\text{anchor}), \qquad \boldsymbol{p} \in [0,1]^M, \quad p_i \equiv p_i(\text{anchor}). \tag{9}$$

For multiple anchors (used in several workflows below), EAGGL generalizes $\boldsymbol{q}$ and $\boldsymbol{p}$ to matrices $Q \in [0,1]^{N\times U}$ and $P \in [0,1]^{M\times U}$ with one column per anchor index u. Internally, EAGGL appends an additional "union" (any-anchor) column to each of Q and P:

$$Q_{\bullet,\text{any}} := 1 - \prod_{u=1}^{U} \left(1 - Q_{\bullet u}\right), \qquad P_{\bullet,\text{any}} := 1 - \prod_{u=1}^{U} \left(1 - P_{\bullet u}\right), \tag{10}$$

and uses the augmented anchor set $U' = U + 1$ when constructing weights.

The weight matrix used by the bNMF objective is then

$$S \;:=\; QP^\top \;\in\; \mathbb{R}_+^{N\times M}, \qquad S_{ji} \;=\; \sum_{u=1}^{U'} Q_{ju} P_{iu}. \tag{11}$$

In the single-anchor setting ($U = 1$), Eq. 11 reduces to a scaled outer product $S_{ji} \propto q_j p_i$. This weighting emphasizes reconstruction accuracy for matrix entries corresponding to genes and gene sets with high anchor relevance.

Gaussian likelihood and ARD prior. EAGGL uses a Bayesian NMF with a weighted Gaussian (L2) reconstruction term and an automatic relevance determination (ARD) prior to prune the effective number of factors. The factorization seeks non-negative matrices

$$W \in \mathbb{R}_+^{N\times K}, \qquad H \in \mathbb{R}_+^{K\times M}, \qquad V \approx WH. \tag{12}$$

The negative log likelihood term implemented in EAGGL is

$$\mathcal{L}_{\text{like}}(W,H) \;=\; \frac{1}{2} \sum_{j=1}^{N} \sum_{i=1}^{M} S_{ji} \left(V_{ji} - (WH)_{ji}\right)^2. \tag{13}$$

ARD introduces per-factor shrinkage weights $\lambda_k > 0$ that penalize the squared norms of $W_{\bullet k}$ and $H_{k\bullet}$:

$$\mathcal{L}_{\mathrm{ARD}}(W, H, \boldsymbol{\lambda}) \;=\; \phi \sum_{k=1}^{K} \left[\frac{\frac{1}{2}\left(\|W_{\bullet k}\|_2^2 + \|H_{k\bullet}\|_2^2\right) + b_0}{\lambda_k} + C \log \lambda_k \right], \tag{14}$$

where $\phi > 0$ controls overall shrinkage strength and (b_0, C) are scalars derived from the matrix dimensions and an ARD hyperparameter a_0. In the current implementation:

$$\phi \leftarrow \phi_{\mathrm{CLI}} \cdot \mathrm{sd}(V)^2, \tag{15}$$

$$C := \frac{N+M}{2} + a_0 + 1, \tag{16}$$

$$b_0 := \frac{3.14\,(a_0 - 1)\;\mathrm{E}[V]}{2K_0}, \tag{17}$$

where K_0 is set to the initial requested K.

The total objective reported as "evidence" in the code is

$$\mathcal{L}(W, H, \boldsymbol{\lambda}) \;=\; \mathcal{L}_{\mathrm{like}}(W, H) + \mathcal{L}_{\mathrm{ARD}}(W, H, \boldsymbol{\lambda}). \tag{18}$$

Non-negativity and preprocessing. EAGGL enforces non-negativity by construction (multiplicative updates) and by preprocessing V as

$$V \leftarrow V - \min(V), \tag{19}$$

and by zeroing negative entries before factorization (for inputs that may contain signed values). During factorization, EAGGL additionally caps entries of W and H to lie in $[0, 1]$ (without requiring row or column sums to equal 1).

Multiplicative updates. EAGGL uses multiplicative updates consistent with weighted NMF and L2-ARD regularization. Let $V_{\mathrm{ap}} := WH$ and let ε be a small constant. Then the updates implemented are:

$$H_{ki} \leftarrow H_{ki} \cdot \frac{\left(W^\top (V \odot S)\right)_{ki}}{\left(W^\top (V_{\mathrm{ap}} \odot S)\right)_{ki} + \phi\, H_{ki} \lambda_k^{-1} + \varepsilon}, \tag{20}$$

$$W_{jk} \leftarrow W_{jk} \cdot \frac{\left((V \odot S) H^\top\right)_{jk}}{\left((V_{\mathrm{ap}} \odot S) H^\top\right)_{jk} + \phi\, W_{jk} \lambda_k^{-1} + \varepsilon}, \tag{21}$$

where $\odot$ denotes elementwise multiplication. After each update, values are clipped to be non-negative (and, in EAGGL's default factoring path, clipped to be at most 1).

The ARD weights are updated as

$$\lambda_k \leftarrow \frac{\frac{1}{2}\left(\|W_{\bullet k}\|_2^2 + \|H_{k\bullet}\|_2^2\right) + b_0}{C}. \tag{22}$$

Iterations continue until the relative change in $\boldsymbol{\lambda}$ drops below a tolerance or a maximum iteration count is reached.

Factor filtering. After convergence, EAGGL filters out weak factors using a combination of thresholds: (i) λ_k must exceed a minimum (set via `--min-lambda-threshold`), (ii) the total mass of gene and gene-set loadings in factor k must exceed the same threshold, and (iii) factors with extremely small peak gene-set loading relative to the overall maximum are dropped. The remaining factors are later reordered by a post-hoc relevance score (Section 1.3).

1.3 Post-processing: trait relevance and projection

EAGGL performs two post-processing operations after learning a factorization on a subset of genes and gene sets: (i) it computes trait relevance quantities for each factor, and (ii) it projects factor loadings to additional entities (genes, gene sets, and traits/phenotypes) when corresponding PIGEAN data are available.

1.3.1 Anchor-specific and any-trait relevance scores

Let $F \in \mathbb{R}_+^{M\times K}$ denote the gene-by-factor loading matrix after factorization and projection (in code: `exp_gene_factors` when factoring gene sets by genes, or `exp_pheno_factors` when factoring gene sets by phenotypes). Let $P \in [0,1]^{M\times U}$ be the matrix of gene (or phenotype) relevance probabilities for the U anchors.

Anchor relevance. For each anchor u, EAGGL computes a non-negative coefficient vector $\boldsymbol{r}^{(u)} \in \mathbb{R}_+^K$ by solving a non-negative least squares (NNLS) projection:

$$\boldsymbol{r}^{(u)} \;=\; \arg\min_{\boldsymbol{r}\geq 0} \|P_{\bullet u} - F\boldsymbol{r}\|_2^2\,, \tag{23}$$

implemented via multiplicative updates for NNLS. The resulting matrix $R_{\text{anchor}} \in \mathbb{R}_+^{K\times U}$ is written as the per-anchor `relevance` column in anchor-specific outputs.

Any-anchor relevance. EAGGL additionally defines an "any" relevance vector across anchors as the probability that a gene (or phenotype) is relevant to *at least one* anchor:

$$P_{\bullet,\text{any}} := 1 - \prod_{u=1}^{U} (1 - P_{\bullet u})\,. \tag{24}$$

It then computes an NNLS projection analogous to Eq. 23,

$$\boldsymbol{r}^{(\text{any})} \;=\; \arg\min_{\boldsymbol{r}\geq 0} \|P_{\bullet,\text{any}} - F\boldsymbol{r}\|_2^2\,, \tag{25}$$

and uses the resulting coefficients as `any_relevance` (in `factors.out`).

1.3.2 Projection of loadings to all genes, gene sets, and traits

Factorization is typically run on a filtered subset of genes and gene sets (based on thresholds on p_i and q_j, and optional pruning for correlation/size). EAGGL then imputes factor loadings for the remaining entities by holding one side of the factorization fixed and updating the other with multiplicative updates that mirror Eq. 20.

Projection with fixed W. Given a learned basis $W \in \mathbb{R}_+^{N\times K}$ and a new matrix $V_{\text{new}} \in \mathbb{R}_+^{N\times M_{\text{new}}}$ (same row entities, new columns), EAGGL estimates loadings $H_{\text{new}} \in \mathbb{R}_+^{K\times M_{\text{new}}}$ by minimizing the weighted L2 objective with fixed W:

$$\min_{H_{\text{new}}\geq 0} \;\frac{1}{2}\,\|(V_{\text{new}} - WH_{\text{new}}) \odot S_{\text{new}}\|_F^2 \;+\; \phi \sum_{k=1}^{K} \lambda_k^{-1}\,\|(H_{\text{new}})_{k\bullet}\|_2^2\,, \tag{26}$$

where S_{new} is constructed from the corresponding row and column probability matrices exactly as in Eq. 11. The update rule matches Eq. 20 with W fixed.

Genes. After factoring gene sets by genes (the default single-phenotype anchoring case), EAGGL projects gene loadings to all genes using $W \leftarrow$ learned gene-set factors and $V_{\text{new}} \leftarrow$ the full gene-set-by-gene matrix derived from `X`.

Traits/phenotypes. If PIGEAN PheWAS outputs are available (either gene PheWAS and/or gene-set PheWAS), EAGGL constructs a gene-set-by-phenotype matrix from those statistics and projects phenotype loadings using Eq. 26. This produces factor loadings for all phenotypes with available PIGEAN data, enabling interpretation of mechanisms beyond the anchor trait.

Gene sets. EAGGL also projects gene-set loadings to all gene sets by swapping roles of rows and columns (effectively using a fixed gene- or phenotype-factor matrix) and projecting gene sets onto the learned factors.

Factor ordering and sparsification. After relevance computation and projection, factors are sorted by decreasing any-anchor relevance, and very small entries in the loading matrices are set to zero (relative to the global maximum loading).

1.4 Alternative factor workflows and implicit models

EAGGL supports a family of related workflows that differ in the choice of anchor, the matrix being factored, and the probability weights used to emphasize anchor-relevant entries. We describe these workflows as F1–F9 following the repository documentation; F1 is the default single-phenotype anchoring described above, and F2–F9 are alternative matrix constructions.

1.4.1 F2: Gene-list-as-phenotype anchoring

Implicit biological model. A gene list (e.g. positive controls for a disease or pathway) is treated as an observable proxy for trait relevance, analogous to the binary d_i in the single-anchor model. Mechanisms are assumed to explain co-membership patterns between the gene list and gene-set annotations.

Calculation. EAGGL first obtains gene-level probabilities and gene-set effect summaries for predicting membership in the input list (via the same upstream pipeline used to obtain anchor probabilities for a phenotype). It then runs the same gene-set-by-gene factorization as in F1, with the gene list acting as the anchor used to define p_i and q_j.

Approximation and limitations. Because the anchor is not a phenotype with genetic association evidence but an input set, the meaning of p_i becomes "probability gene belongs to the list" rather than "probability gene is trait-relevant". Downstream projections to phenotypes remain possible if PheWAS summaries are available, but interpretation depends on how faithfully the gene list represents a coherent biological process.

1.4.2 F3: Single phenotype anchoring with projection from PheWAS

Implicit biological model. This workflow assumes the same mechanism structure as the single-anchor model, but additionally assumes that the inferred mechanisms can be re-used to explain gene and gene-set associations for other phenotypes.

Calculation. EAGGL runs F1 to learn gene-set and gene loadings. Given external PIGEAN PheWAS summaries, it projects phenotypes onto the learned factors using Eq. 26, producing phenotype-by-factor loadings for all phenotypes in the PheWAS files.

Approximation and limitations. Projection assumes that the factor basis learned from the anchor trait is adequate for other phenotypes; phenotypes whose mechanism structure differs substantially may be poorly represented.

1.4.3 F4: Multi-phenotype anchoring

Implicit biological model. Multiple anchor phenotypes $\{t_u\}_{u=1}^{U}$ are assumed to share some subset of mechanisms. Genes and gene sets that are relevant to multiple anchors are treated as stronger evidence for shared mechanisms.

Calculation. Using gene and gene-set PheWAS statistics, EAGGL constructs $P \in [0,1]^{M \times U}$ and $Q \in [0,1]^{N \times U}$ for the selected anchor phenotypes, and factors the same gene-set-by-gene matrix V as in F1. The anchors enter through the weight matrix $S = QP^\top$ (Eq. 11), which aggregates anchor-specific outer products and an additional union column.

Approximation and limitations. Compared to an explicit tensor factorization that preserves a separate slice per phenotype, the implemented weight aggregation compresses multi-phenotype information into a single weighted objective. This can blur anchor-specific differences: mechanisms that are strong for one anchor but weak for others can still influence the weights through S.

1.4.4 F5: Any-phenotype anchoring

Implicit biological model. An "uber" phenotype is defined as relevance to *any* phenotype in a provided PheWAS panel. Mechanisms are inferred to explain gene and gene-set patterns that are collectively associated with at least one phenotype.

Calculation. EAGGL sets the anchor phenotype mask to include all phenotypes in the PheWAS input, then collapses probabilities to a single column using Eq. 24 (applied to both genes and gene sets). It then runs the F1-style factorization on $V = X^\top$ using the collapsed probabilities.

Approximation and limitations. If many phenotypes are provided, the any-phenotype probability can saturate for broadly associated genes or annotations, reducing contrast in the weighting scheme. Optional pruning of genes, gene sets, and phenotypes is used to control matrix size and correlation structure.

1.4.5 F6: Single-gene anchoring

Implicit biological model. A single gene is treated as the anchor, and mechanisms are defined by patterns of phenotypes that co-associate with that gene and with gene sets. This can be interpreted as learning phenotype modules connected to the anchor gene through latent mechanisms.

Calculation. EAGGL factors a gene-set-by-phenotype matrix constructed from gene-set PheWAS statistics (using uncorrected betas when available). Column weights are derived from the anchor gene's PheWAS association probabilities across phenotypes. Gene-set weights are not required in this workflow and default to uniform.

Approximation and limitations. Because the factored matrix is gene-set-by-phenotype rather than gene-set-by-gene, the gene loading side becomes a phenotype loading side. Interpretation of f^g is therefore replaced by phenotype loadings, and gene-level loadings are obtained only through downstream projection steps.

1.4.6 F7: Multi-gene anchoring

Implicit biological model. Multiple anchor genes are assumed to implicate a shared set of latent mechanisms, which in turn connect to phenotypes and gene sets. Phenotypes associated with any of the anchor genes provide evidence for these shared mechanisms.

Calculation. EAGGL uses gene PheWAS summaries to construct phenotype probabilities for each anchor gene, then aggregates these through the weight matrix $S = QP^\top$ as in Eq. 11 (with uniform gene-set weights by default). The factored matrix is again gene-set-by-phenotype (as in F6).

Optional expansion heuristics can add gene sets to the matrix based on enrichment or overlap with the anchor gene set (e.g. by enrichment p-value or fraction overlap), increasing coverage before factorization.

Approximation and limitations. Multi-gene anchoring compresses a potentially heterogeneous set of anchors into a single weighted objective. If anchor genes reflect distinct biology, the resulting factors can blend mechanisms. Expansion heuristics increase sensitivity but can also introduce redundancy or off-target gene sets.

1.4.7 F8: Any-gene anchoring

Implicit biological model. An "uber" gene anchor is defined as association to *any* gene in the provided gene panel (typically all genes with PheWAS summaries). This workflow aims to capture broad phenotype–gene-set structure driven by genetic associations.

Calculation. EAGGL sets the anchor gene mask to include all genes, constructs phenotype probabilities for each gene, and then collapses to a single probability per phenotype using Eq. 24 across genes. It then factors the gene-set-by-phenotype matrix as in F6.

Approximation and limitations. As in any-phenotype anchoring, union probabilities can saturate when many genes are included, reducing effective weighting contrast. Pruning controls are typically required to make the matrix tractable and to reduce correlation.

1.4.8 F9: Gene-set anchoring

Implicit biological model. A gene set is treated as the anchor object. Phenotypes are weighted by their association to the anchor gene set, and other gene sets are weighted by their relevance

under the anchor gene set model. Mechanisms are then inferred as latent modules that explain gene-set-by-phenotype association structure near the anchor.

Calculation. EAGGL factors a gene-set-by-phenotype matrix built from gene-set PheWAS statistics. Phenotype weights are derived from a PheWAS run that evaluates association between phenotypes and input gene scores (producing a phenotype-level evidence vector used as the anchor column weights). Gene-set weights are derived from the available gene-set evidence summaries. The weighted bNMF proceeds as in Section 1.3, with S built from these gene-set and phenotype weights.

Approximation and limitations. This workflow depends on the quality and calibration of phenotype weights produced by the gene-set-to-phenotype PheWAS step. If those weights are noisy or biased, they can dominate the factorization and lead to unstable or hard-to-interpret mechanisms.

1.5 Future directions

The draft above documents the models and equations as implemented in the current EAGGL codebase. Several modeling and implementation issues are known and suggest concrete future work:

- **Explicit probabilistic linkage to the single-anchor model.** Eq. 7 motivates factoring a weighted matrix $p_i q_j X_{ij}$. The current implementation instead emphasizes anchor-relevant entries through a weighted loss (Eq. 13) built from S. A useful extension would be to derive and implement a generative model in which the observed matrix is *explicitly* $p_i q_j X_{ij}$ (or an empirically calibrated transform thereof), and to evaluate whether this improves interpretability.

- **Multi-anchor tensor or collective factorization.** Multi-phenotype and multi-gene workflows currently aggregate anchors through $S = QP^\top$. An extension would preserve per-anchor structure explicitly (e.g. a tensor factorization with a shared gene-set-by-gene basis and anchor-specific factor weights) to better capture anchor-specific mechanisms.

- **Multi-start optimization and stability diagnostics.** The bNMF optimization is nonconvex; repeated random initializations and stability-based model selection would improve robustness. The current CLI encourages re-running, but multi-start selection is not implemented as a first-class feature.

- **Hyperparameter surface and unused knobs.** The bNMF implementation uses ϕ and a_0 (via `--phi` and `--alpha0`) in the ARD objective, while other CLI knobs (e.g. `--beta0`) are currently not consumed by the factorization routine. Auditing and aligning the exposed parameterization with the implemented objective would reduce confusion and enable principled tuning.

- **Calibration of probability transforms.** Gene and gene-set probabilities are computed via a logistic transform with a shared background prior. A more principled calibration could propagate uncertainty from PIGEAN, use trait-specific priors, or learn monotone calibrations that improve downstream interpretability.

- **Signed or antagonistic effects.** Some annotation and PheWAS inputs can be signed. EAGGL currently enforces non-negativity by zeroing negatives and shifting matrices; extending to signed factorizations (or separate positive/negative channels) could capture antagonistic mechanisms.
- **Improved relevance scoring.** Factor relevance is computed via NNLS projection of gene (or phenotype) probabilities onto factor loadings. Alternative definitions (e.g. probabilistic relevance models, enrichment-based scores, or Bayesian regression) could better align relevance with the mechanistic model and provide uncertainty estimates.